

ORIGINAL RESEARCH

Exploration of the Interrelationship Between Serum Uric Acid, Gout, and Cardiac, Renal, and Metabolic Conditions in Middle Aged and Older People

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BACKGROUND: Cardiac, renal, and metabolic (CRM) conditions are major causes of morbidity and mortality globally. This study aims to explore the relationship between serum uric acid (SUA), hyperuricemia, gout, and CRM conditions in middle-aged and elderly populations.

METHODS: Sample 1 included participants from CHARLS (China Health and Retirement Longitudinal Study, n=9341), and Sample 2 from NHANES (National Health and Nutrition Examination Survey, unweighted n=17 913; weighted n=115 646 390). Ordinal logistic regression, Cox regression, and restricted cubic spline analyses were used to assess the relationship between SUA, hyperuricemia, gout, and CRM conditions. A 2-sample Mendelian randomization analysis was conducted to explore causal associations between SUA and CRM conditions.

RESULTS: In both samples, SUA, hyperuricemia, and gout were positively correlated with the risk of CRM conditions. Among participants with 3 or ≥1 CRM condition(s), SUA, asymptomatic hyperuricemia, and gout with poorly controlled hyperuricemia showed significant positive associations with all-cause mortality, whereas these associations were not observed in patients with gout with normal SUA levels. The restricted cubic spline analysis revealed a positive relationship between SUA levels and the risk of all-cause mortality in participants with ≥1 CRM condition(s), demonstrating a nonlinear dose–response relationship across both samples (*P* for nonlinearity <0.05). Mendelian randomization analysis indicated that SUA was causally associated with cardiovascular disease, chronic kidney disease, and diabetes.

CONCLUSIONS: Hyperuricemia and gout are strong predictors of increased prevalence and mortality of CRM conditions, emphasizing the importance of managing hyperuricemia and gout in these patients.

Key Words: cardiac, renal, and metabolic conditions ■ gout ■ hyperuricemia ■ Mendelian randomization ■ serum uric acid

In recent years, the American Heart Association introduced the concept of “cardiac, renal, and metabolic (CRM) conditions,” describing the complex interactions and pathophysiological links between the

cardiovascular system, renal system, and metabolic risk factors, which ultimately lead to multiorgan dysfunction and complex clinical manifestations.^{1,2} CRM conditions have a high global prevalence and are

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CLINICAL PERSPECTIVE

What Is New?

- This study establishes a clear link between serum uric acid, hyperuricemia, and gout with the risk of cardiac, renal, and metabolic (CRM) conditions, highlighting their impact on all-cause mortality in middle-aged and elderly populations.

What Are the Clinical Implications?

- The findings suggest that clinicians should consider hyperuricemia and gout as key factors in the management of patients with cardiac, renal, and metabolic conditions, as these conditions are associated with increased morbidity and mortality.

Nonstandard Abbreviations and Acronyms

CHARLS	China Health and Retirement Longitudinal Study
CRM	cardiac, renal, and metabolic
MR	Mendelian randomization
NHANES	National Health and Nutrition Examination Survey
SNV	single-nucleotide variant
SUA	serum uric acid

among the leading causes of morbidity and mortality, accounting for approximately one third of all deaths worldwide.³ The interconnected nature of these conditions suggests that dysfunction in one system can trigger or worsen dysfunction in others. Among adults in the United States, the estimated prevalence of individual CRM conditions is approximately 9% to 11% for cardiovascular disease (CVD), 15% for chronic kidney disease (CKD), and 13% for diabetes, with >1 in 4 individuals having at least 1 CRM condition and nearly 1 in 10 experiencing overlapping CRM conditions.¹ Given the significant cumulative associations between coexisting CRM conditions and adverse outcomes, including mortality, disability, and economic burden,⁴ optimizing care pathways to mitigate their impact is crucial. Regular exercise, along with well-controlled blood pressure, serum glucose, and lipid levels, can significantly improve the prognosis of patients with CRM conditions and their widely recognized risk factors, including dysglycemia, dyslipidemia, hypertension, and obesity.^{5,6}

Uric acid is the final product of purine metabolism, primarily synthesized in the liver and excreted

through the renal tract and intestines.⁷ At physiological concentrations, circulating uric acid acts as an antioxidant and can scavenge reactive oxygen species.⁸ Hyperuricemia, traditionally associated with gout, is now recognized as a key contributor to cardiovascular and metabolic diseases, including hypertension, CVD, CKD, diabetes, hypertriglyceridemia, and metabolic syndrome.⁹ Elevated serum uric acid (SUA) is often linked with insulin resistance, hyperglycemia, and dyslipidemia, suggesting its involvement in the metabolic environment underlying CRM conditions.¹⁰ The pathogenic effects of SUA are largely mediated through its proinflammatory and prooxidative properties. Elevated SUA induces oxidative stress by activating NADPH oxidase, leading to increased production of reactive oxygen species.¹¹ It also triggers inflammatory responses by upregulating mediators such as monocyte chemoattractant protein-1, C-reactive protein, and promoting platelet activation.¹² These processes result in endothelial dysfunction, vascular remodeling, and insulin resistance, which together contribute to the development and progression of CVD, CKD, and diabetes. Additionally, SUA may indirectly influence CRM conditions through its association with metabolic risk factors such as obesity and dyslipidemia, further exacerbating disease pathways. These mechanisms highlight SUA's dual role as both a direct pathogen and an indirect amplifier of metabolic dysfunction. Due to changes in dietary patterns, improved living conditions, and aging populations, the global prevalence and disease burden of hyperuricemia and gout have been steadily increasing.¹³ However, despite the growing understanding of SUA's contribution to CRM pathogenesis, its interaction with gout, other metabolic factors, and CRM outcomes—particularly in middle-aged and older populations—remains underexplored.

This study aims to investigate these relationships using data from 2 large databases, CHARLS (China Health and Retirement Longitudinal Study) and NHANES (National Health and Nutrition Examination Survey), and to assess the causal association between SUA and CRM conditions through Mendelian randomization (MR) analysis.

METHODS

Data Source and Study Population

This study used 2 large samples from the CHARLS and NHANES databases. All data and materials have been made publicly available at the CHARLS database (<https://charls.pku.edu.cn/>) and NHANES database (https://www.cdc.gov/nchs/nhanes/nhanes_questionnaires.htm). The study was approved by the ethics committee of Peking University (IRB00001052–11015) and the National Center for Health Statistics' Research

Ethics Review Board. Written informed consent was obtained from all participants.

Sample 1 included participants from the CHARLS database. CHARLS is a national population-based survey of Chinese adults aged 45 years or older, conducted by the National School of Development at Peking University. Participants were selected using a 4-stage, stratified, cluster random sampling method from 150 counties across 28 provinces.¹⁴ This study used cross-sectional data from wave 3 of the CHARLS survey, including 9341 eligible participants. [Figure S1](#) shows the participant selection process.

Sample 2 included participants from the NHANES database. NHANES, conducted by the Centers for Disease Control and Prevention, collects data on the health and nutritional status of the US population every 2 years. It selects representative samples of noninstitutionalized civilians.¹⁵ Participants in this analysis were from 6 2-year survey cycles conducted between 2007 and 2018. The participant screening process is shown in [Figure S2](#). A large, nationally representative sample of US adults aged ≥ 45 years was identified (unweighted $n=17\,913$; weighted $n=115\,646\,390$), including 1314 (7.3%) patients with gout.

Study Variables

The variables for both samples included age (years), sex (male, female), body mass index (BMI) categories (Sample 1: <24 kg/m², 24–28 kg/m², ≥ 28 kg/m²; Sample 2: <25 kg/m², 25–30 kg/m², ≥ 30 kg/m²), education level (Sample 1: less than lower secondary, upper secondary and vocational training, tertiary; Sample 2: less than 9th grade, 9th–11th grade, high school graduate, college or associate degree, college graduate or above), marital status (Sample 1: married/living with partner, widowed/divorced/separated; Sample 2: married/living with partner, widowed/divorced/separated, never married), estimated glomerular filtration rate (calculated using the Chronic Kidney Disease Epidemiology Collaboration equation),¹⁶ low-density lipoprotein cholesterol (mmol/L), triglycerides (mmol/L), SUA (mg/dL), and smoking history (yes, no). The history of hypertension (yes, no), CVD (yes, no), CKD (yes, no), and diabetes (yes, no) were also included. Additionally, Sample 1 included residence (rural or urban) as a variable. Sample 2 included additional variables such as race or ethnicity (Mexican American, other Hispanic, non-Hispanic White, non-Hispanic Black, other race or ethnicity, such as Non-Hispanic Asian and Multi-Racial), poverty:income ratio, and history of gout (yes, no). Poverty:income ratio was used to define income, with poverty defined as a poverty:income ratio <0.99 .

In CHARLS, blood samples were collected from participants in the Venous Blood-Based Biomarkers in the China Health study after an overnight fast by

uniformly trained staff from the Chinese Center for Disease Control and Prevention, following standard protocols. SUA levels were measured using the UA Plus method.¹⁷ In NHANES, SUA levels were measured using the DxC800 synchron with a timed end point method.¹⁸

Regarding survival status, the latest release of CHARLS Wave 5 (2020) data provided survival status (alive or deceased) for participants from Wave 3. However, specific dates of death were not provided for deceased participants. For these individuals, survival time was estimated by calculating the median time from T2 to T3, combined with the time from baseline to T2.

The National Center for Health Statistics linked data from various surveys with death certificate records from the National Death Index, with follow-up through NHANES 1999 to 2018. The Linked Mortality Files were updated with mortality data through December 31, 2019. In NHANES, follow-up time is defined as the time from the interview to the date of death or the end of the follow-up period.¹⁹

Definition

Hyperuricemia is defined as an SUA level greater than 7 mg/dL in men and postmenopausal women, or >6 mg/dL in premenopausal women.²⁰ In this study, hyperuricemia was defined as an SUA level >7 mg/dL for all middle-aged and older participants.²¹ Asymptomatic hyperuricemia was defined as an SUA level >7 mg/dL, with individuals answering “No” to the question, “Has a doctor ever told you that you had gout?” Gout was defined as answering “Yes” to the question, “Has a doctor ever told you that you had gout?” CKD was defined as an estimated glomerular filtration rate <60 or the presence of albuminuria (urine albumin-creatinine ratio ≥ 30 mg/g).²² Diabetes was defined as either a self-reported diagnosis by a health care professional or a glycated hemoglobin level of at least 6.5%. CVD was defined as self-reported diagnosis of 1 or more of the following by a health care professional: heart failure, coronary heart disease, myocardial infarction, or stroke.¹

Genome-Wide Association Study Sources and Selection of Instruments for Mendelian Randomization

[Figure S3](#) presents a schematic diagram of the 2-sample MR analysis. Summary data for SUA variants were obtained from the UK Biobank database, which enrolled 343 836 Europeans,²³ with a mean SUA level of 309.43 ± 80.39 $\mu\text{mol/L}$. Summary data for CVD (185 353 cases and 191 924 controls), CKD (9073 cases and 363 177 controls), and diabetes (38 657

cases and 310 131 controls) variants were obtained from the FinnGen database (r9).²⁴

An MR study must meet 3 key assumptions. First, the genetic instrument for exposure must be strongly associated with the trait of interest. Second, the selected genetic variants should not affect the outcome through any other pathways independent of the exposure. Finally, the genetic variants must not be associated with any confounders of SUA or CRM conditions.

The genome-wide significance threshold of $P < 5 \times 10^{-8}$ was applied to identify genetic variants strongly associated with SUA. Single-nucleotide variants (SNVs) with linkage disequilibrium ($r^2 < 0.001$, 10 000 kb) were excluded. A total of 268 SNVs strongly associated with SUA were selected for causality analysis, with the proportion of variance in SUA levels explained by the SNVs estimated at 8.53%. Detailed information regarding data downloading and screening is provided in Table S1 and S2. All selected SNVs had an F -statistic > 10 .

We used the LD Link web tool (<https://ldlink.nih.gov/?tab=home>) to assess the associations of the 268 SNVs with confounding factors and outcomes. SNVs associated with confounding factors (eg, BMI, renal function, educational attainment, hypertension, smoking, triglycerides) and outcomes (CVD, CKD, and diabetes) were excluded, leaving a total of 176 SNVs. Of these 176 SNVs, 155 were directly matched in the summary data of SNV–outcome genome-wide association study. We then excluded palindromic SNVs, leaving 148 SNVs for SUA on CVD, 149 for SUA on CKD, and 149 for SUA on diabetes as instrumental variables.

Statistical Analysis

Continuous variables were expressed as mean $\pm$ SD. Student's t test or Mann–Whitney U test was used to assess differences in continuous variables between groups, and the chi-square test was used for categorical variables. To account for the complex sampling design and ensure nationally representative estimates, all analyses for Sample 2 were adjusted for survey design and weighting variables.²⁵ Multinomial ordinal logistic regression models were used to estimate the association between SUA, hyperuricemia, and the number of CRM conditions. Hazard ratios (HRs) and 95% CIs were calculated using multivariable Cox proportional hazards regression models to assess the associations of SUA levels and hyperuricemia with all-cause mortality risk, adjusting for potential confounders. Age, sex, and race were adjusted for in model 1, and covariates with significant differences between the 2 groups were further adjusted for in model 2. Due to sample size limitations and the number of outcome events, Cox regression analysis was limited to participants with 0

CRM or ≥ 1 CRM in Sample 1. In Sample 2, we also analyzed the relationship between gout and CRM conditions, illustrating overall survival time (in months) using Kaplan–Meier curves, which were compared with a log-rank test based on the number of CRM conditions. Additionally, restricted cubic spline analysis was performed to examine the dose–response relationship between SUA levels and all-cause mortality. This relationship was further assessed through stratified analyses by sex (male, female), age groups (45–64, 65 years and older), and BMI categories (Sample 1: < 24 kg/m², 24–28 kg/m², ≥ 28 kg/m²; Sample 2: < 25 kg/m², 25–30 kg/m², ≥ 30 kg/m²).

For the 2-sample MR analysis, the inverse variance weighted method was the primary approach used to assess causal associations between SUA and CRM conditions. The MR Egger method and weighted median method were also employed. Cochran's Q test was used to estimate heterogeneity between SNVs. The intercept of MR-Egger was used to assess pleiotropy of the genetic variants. We used MR Pleiotropy Residual Sum and Outlier (MR-PRESSO) to evaluate the extent of horizontal pleiotropy. Outlier variants identified by the MR-PRESSO outlier test were removed step by step to reduce heterogeneity and the effect of horizontal pleiotropy. Leave-one-out analysis was performed to determine whether the causal estimate was influenced by a single SNV. This study adhered to the Declaration of Helsinki. All statistical analyses were performed using R 4.4.1 and SPSS 23. A 2-sided P value < 0.05 was considered statistically significant.

RESULTS

Clinical Characteristics of the Study Subjects According to Serum Uric Acid Levels

Table S3 presents the clinical characteristics of participants from Sample 1 (CHARLS). Compared with participants with normal UA levels, those with hyperuricemia were older (62.93 [9.64] versus 61.24 [9.18]), predominantly male (79.8% versus 43.9%), had higher BMI and triglyceride levels, and exhibited a greater prevalence of smoking (69.1% versus 41.6%), hypertension (51.8% versus 32.5%), CVD (23.4% versus 19.8%), diabetes (28.7% versus 21.2%), and CKD (28.3% versus 13.1%), all with P values < 0.05 .

Table S4 presents the clinical characteristics of participants from Sample 2 (NHANES). Participants with hyperuricemia were older (62.49 [10.89] versus 60.16 [10.41]), predominantly male (69.8% versus 43.8%), had higher BMI and triglyceride levels, and exhibited a greater prevalence of smoking (54.4% versus 47.0%), poverty (13.2% versus 10.6%), hypertension (67.0%

versus 43.6%), CVD (22.9% versus 11.8%), CKD (42.1% versus 18.4%), and diabetes (26.2% versus 17.5%). In both samples, participants with hyperuricemia exhibited lower estimated glomerular filtration rate ($P<0.01$) and a higher number of CRM conditions ($P<0.01$).

The clinical characteristics according to SUA levels for patients with at least 1 CRM condition from the CHARLS and NHANES databases are presented in Tables S5 and S6. Table S7 shows the mortality rate of patients with different CRM conditions. As the number of CRM conditions increases, the mortality rate during follow-up rises progressively, with participants having 3 CRM conditions showing the highest mortality rate (18.1% and 45.9%). Participants with at least 1 CRM condition exhibited a mortality rate of 23.6% during follow-up.

Observational Correlation Between Hyperuricemia, Gout, and CRM Conditions

The results of ordinal logistic regression models evaluating the relationship between SUA and CRM conditions are shown in Tables 1 and 2. In the fully adjusted model 2, significant positive associations were found between SUA (CHARLS, odds ratio [OR], 1.078 [95% CI, 1.043–1.114]; NHANES, OR, 1.155 [95% CI, 1.154–1.156] per mg/dL increase) and hyperuricemia (CHARLS, OR, 1.483 [95% CI, 1.275–1.724]; NHANES, OR, 1.646 [95% CI, 1.644–1.648]) with a higher number of CRM conditions among adults aged ≥ 45 years. We also assessed the association between gout and CRM conditions in Sample 2. In the adjusted model 2, gout was associated with an increased likelihood (OR, 1.675 [95% CI, 1.673–1.678]) of having a higher number of CRM conditions compared with participants without gout.

Table 1. Odds Ratios of Ordinal Logistic Regression for the Relationship Among Different Serum Uric Acid Levels and CRM Conditions (CHARLS 2015)

	CRM conditions		
	Crude	Model 1	Model 2
Serum uric acid (mg/dL)	1.163 (1.132–1.195)**	1.197 (1.096–1.308)**	1.078 (1.043–1.114)**
Normal uric acid	1.0 (ref)	1.0 (ref)	1.0 (ref)
Hyperuricemia	1.881 (1.637–2.162)**	1.962 (1.700–2.264)**	1.483 (1.275–1.724)**

CHARLS indicates China Health and Retirement Longitudinal Study; and CRM, cardiac, renal, and metabolic.

Model 1: Adjusted for age and sex.

Model 2: Adjusted for age, sex, residence, education, body mass index, hypertension, smoking status, and triglyceride.

* $P<0.05$.

** $P<0.01$.

Table 2. Weighted Odds Ratios of Ordinal Logistic Regression for the Relationship Among Different Serum Uric Acid Levels, Gout, and CRM Conditions (NHANES, 2007–2018)

	CRM conditions		
	Crude	Model 1	Model 2
Serum uric acid (mg/dL)	1.334 (1.333–1.335)**	1.320 (1.319–1.321)**	1.155 (1.154–1.156)**
Normal control	1.0 (ref)	1.0 (ref)	1.0 (ref)
Asymptomatic Hyperuricemia	2.454 (2.451–2.457)**	2.238 (2.235–2.241)**	1.646 (1.644–1.648)**
Nongout	1.0 (ref)	1.0 (ref)	1.0 (ref)
Gout	2.894 (2.890–2.898)**	2.270 (2.266–2.273)**	1.675 (1.673–1.678)**

CRM indicates cardiac, renal, and metabolic; and NHANES, National Health and Nutrition Examination Survey.

Model 1: Adjusted for age, sex, and race.

Model 2: Adjusted for age, sex, race, education, poverty, body mass index, hypertension, smoking status, and triglyceride.

* $P<0.05$.

** $P<0.01$.

For the CHARLS database sample, the association between hyperuricemia and CRM conditions in model 2 for all participants was largely consistent with results stratified by age, sex, and BMI (Table S8). However, the association between SUA and CRM conditions became insignificant in the age group of 45 to 64 years, the male group, and the group with obesity. For the NHANES database sample, the findings for the association between SUA, asymptomatic hyperuricemia, and gout with CRM conditions in model 2 for all participants were largely consistent with results stratified by age, sex, and BMI (Table S9).

The Kaplan–Meier survival curves for unadjusted all-cause mortality (months) of participants from the NHANES database with varying numbers of CRM conditions are shown in Figure 1. Participants with a higher number of CRM conditions exhibited significantly greater all-cause mortality (Log-rank $P<0.0001$). Within the median follow-up period of 6.17 years, the 5-year survival rates for participants with 0, 1, 2, and 3 CRM conditions were 95.9%, 89.5%, 78.4%, and 64.5%, respectively.

The results of Cox proportional hazards models for different numbers of CRM conditions are presented in Tables 3 and 4. Among participants with no CRM conditions, and after adjusting for potential covariates (Model 2), no significant association was observed between SUA, hyperuricemia, and all-cause mortality in both the CHARLS and NHANES samples. In contrast, among participants with ≥ 1 CRM condition, both SUA levels and hyperuricemia showed significant positive associations with all-cause mortality. The fully adjusted HRs for SUA and hyperuricemia in Model 2 were 1.090 (95% CI, 1.012–1.175) and 1.425 (95% CI, 1.070–1.898)

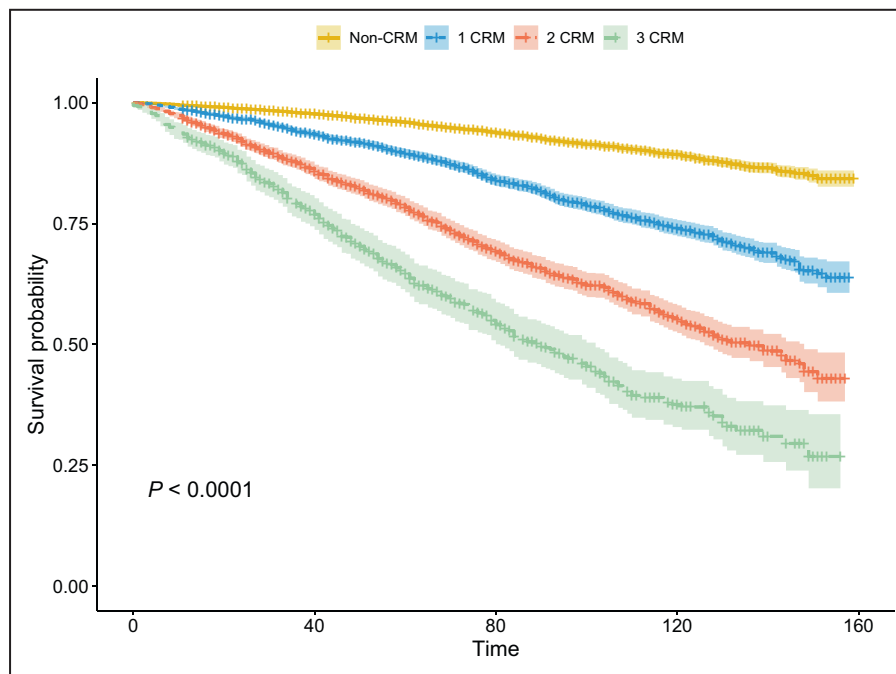


Figure 1. Kaplan–Meier survival curve for all-cause mortality (months) in participants with 0 to 3 CRM conditions.

CRM indicates cardiac, renal, and metabolic.

in the CHARLS sample, and 1.053 (95% CI, 1.006–1.101) and 1.268 (95% CI, 1.091–1.474) in the NHANES sample. Specifically, patients with gout with poorly controlled hyperuricemia had significantly higher all-cause mortality in the groups with 2 (HR, 1.781 [95% CI, 1.069–2.995]), 3 (HR, 2.117 [95% CI, 1.256–3.571]), and ≥ 1 (HR, 1.491 [95% CI, 1.151–1.930]) CRM conditions, whereas no significant association was found

in the 0 and 1 CRM condition groups in the NHANES sample.

In Sample 1, the restricted cubic spline (Figure 2B) showed a positive relationship between SUA and all-cause mortality among patients with ≥ 1 CRM condition ($P < 0.01$, P for nonlinearity < 0.05), with the minimal threshold (lower limit of 95% CIs) of SUA at 7.28 mg/dL. In Sample 2, the restricted cubic spline (Figure 3) demonstrated a positive relationship between SUA and all-cause mortality among patients with 1, 2, 3, and ≥ 1 CRM conditions ($P < 0.05$, P for nonlinearity < 0.05). The minimal threshold (lower limit of 95% CIs) of SUA was 7.67 mg/dL, 7.49 mg/dL, 6.64 mg/dL, and 7.31 mg/dL, respectively.

Stratified analyses by age, sex, and BMI categories were performed for participants with 0 or ≥ 1 CRM condition (Tables S10 and S11). The findings for the association between SUA, hyperuricemia, and gout with all-cause mortality in Model 2 across all participants were largely consistent with results stratified by age, sex, and BMI among participants with 0 CRM conditions. For participants with ≥ 1 CRM condition, similar results were observed in those aged 65 years and above, the female group, and the group with obesity in the NHANES sample.

Table 3. Hazard Ratios (95% CIs) for All-Cause Mortality According to Serum Uric Acid and Hyperuricemia Among Participants With Different CRM Conditions (CHARLS 2015)

	Serum uric acid (mg/dL)	Hyperuricemia [†]
0 CRM (n=5316)		
Crude	1.160 (1.069–1.259)**	1.339 (0.874–2.051)
Model 1	1.020 (0.931–1.118)	0.968 (0.629–1.490)
Model 2	1.035 (0.931–1.150)	0.920 (0.578–1.465)
≥ 1 CRM (n=4025)		
Crude	1.187 (1.120–1.258)**	1.941 (1.510–2.495)**
Model 1	1.076 (1.009–1.148)*	1.443 (1.117–1.865)**
Model 2	1.090 (1.012–1.175)*	1.425 (1.070–1.898)*

CHARLS indicates China Health and Retirement Longitudinal Study; and CRM, cardiac, renal, and metabolic.

Model 1: Adjusted for age and sex.

Model 2: Adjusted for age, sex, residence, education, body mass index, hypertension, smoking status, estimated glomerular filtration rate, and triglyceride.

* $P < 0.05$.

** $P < 0.01$.

[†]Compared with participants without hyperuricemia.

Causal Association Between SUA and CRM Conditions

We further conducted MR analysis to investigate the causal effects of SUA on CVD, CKD, and diabetes. As

Table 4. Weighted Hazard Ratios (95% CIs) for All-Cause Mortality According to Serum Uric Acid, Hyperuricemia, and Gout Among Participants With Different CRM Conditions (NHANES, 2007–2018)

	Serum uric acid (mg/dL)	Asymptomatic hyperuricemia [†]	Gout with normal UA [†]	Gout with poorly-controlled hyperuricemia [†]
0 CRM (n=9662)				
Crude	1.010 (0.938–1.087)	1.263 (0.890–1.792)	1.454 (0.811–2.605)	1.547 (0.687–3.485)
Model 1	0.910 (0.830–0.998)*	1.133 (0.803–1.600)	1.012 (0.571–1.795)	1.092 (0.483–2.469)
Model 2	0.921 (0.833–1.021)	1.159 (0.800–1.680)	1.057 (0.585–1.909)	1.137 (0.478–2.707)
1 CRM (n=5156)				
Crude	1.052 (0.992–1.115)	1.286 (1.018–1.623)*	1.284 (0.878–1.879)	1.304 (0.837–2.031)
Model 1	1.020 (0.956–1.088)	1.242 (0.998–1.546)	1.053 (0.724–1.530)	1.206 (0.804–1.809)
Model 2	1.006 (0.940–1.077)	1.207 (0.972–1.498)	0.974 (0.672–1.411)	1.187 (0.758–1.860)
2 CRM (n=2346)				
Crude	1.130 (1.069–1.194)**	1.400 (1.077–1.817)*	1.206 (0.837–1.737)	1.601 (1.168–2.196)**
Model 1	1.122 (1.055–1.194)**	1.333 (1.022–1.739)*	1.127 (0.780–1.629)	1.640 (1.153–2.333)**
Model 2	1.053 (0.965–1.150)	1.146 (0.791–1.661)	0.821 (0.439–1.538)	1.781 (1.069–2.995)*
3 CRM (n=749)				
Crude	1.138 (1.054–1.229)**	1.762 (1.200–2.587)**	1.423 (0.978–2.072)	2.223 (1.375–3.592)**
Model 1	1.122 (1.043–1.207)**	1.735 (1.216–2.477)**	1.435 (0.926–2.223)	1.992 (1.197–3.316)**
Model 2	1.128 (1.047–1.216)**	1.830 (1.251–2.679)**	1.558 (0.981–2.474)	2.117 (1.256–3.571)**
≥1 CRM (n=8251)				
Crude	1.161 (1.121–1.202)**	1.661 (1.432–1.927)**	1.478 (1.161–1.882)**	1.957 (1.542–2.482)**
Model 1	1.129 (1.082–1.178)**	1.539 (1.323–1.789)**	1.261 (1.009–1.577)*	1.806 (1.418–2.300)**
Model 2	1.053 (1.006–1.101)*	1.268 (1.091–1.474)**	1.116 (0.886–1.407)	1.491 (1.151–1.930)**

CRM indicates cardiac, renal, and metabolic conditions; and NHANES, National Health and Nutrition Examination Survey.

Model 1: Adjusted for age, sex and race.

Model 2: Adjusted for age, sex, race, education, poverty, body mass index, hypertension, smoking status, estimated glomerular filtration rate, and triglyceride.

* $P<0.05$.

** $P<0.01$.

[†]Compared with participants without hyperuricemia or gout.

shown in Table 5, MR results demonstrated that genetically predicted SUA levels were positively associated with CVD, CKD, and diabetes (SUA on CVD: inverse variance weighted OR, 1.109 [95% CI, 1.024–1.201]; $P<0.01$; SUA on CKD: inverse variance weighted OR, 1.441 [95% CI, 1.231–1.688]; $P<0.01$; SUA on diabetes: inverse variance weighted OR, 1.189 [95% CI, 1.075–1.316]; $P<0.01$). The estimated effect sizes of the instrumental variable SNVs of SUA on CVD, CKD, and diabetes are displayed in scatter plots (Figures S4, S5, and S6). MR-Egger regression indicated no notable pleiotropy (SUA on CVD: intercept=−0.00018, $P=0.911$; SUA on CKD: intercept=0.00531, $P=0.106$; SUA on diabetes: intercept=−0.00019, $P=0.927$). The forest plot, leave-one-out analysis, and funnel plot are presented in Figures S7–S15. No potential influential SNVs were found in the leave-one-out analysis. MR-PRESSO analysis was further performed, and after removing outlier SNVs, MR results remained significant (SUA on CVD: MR-PRESSO OR, 1.124 [95% CI, 1.048–1.206]; $P<0.01$; SUA on CKD: MR-PRESSO OR, 1.425 [95% CI, 1.219–1.665]; $P<0.01$; SUA on

diabetes: MR-PRESSO OR, 1.171 [95% CI, 1.063–1.289]; $P<0.01$).

DISCUSSION

This study, for the first time, integrated data from the CHARLS and NHANES databases to systematically investigate the associations between SUA, hyperuricemia, gout, and CRM conditions. Using various statistical approaches and MR analyses, we confirmed the robustness of our findings. The study revealed significant positive associations between SUA, hyperuricemia, gout, and CRM conditions, with higher numbers of CRM conditions correlating with increased all-cause mortality. Dose-response analyses demonstrated a nonlinear positive relationship between SUA levels and all-cause mortality risk. Furthermore, MR confirmed the causal associations of SUA with CVD, CKD, and diabetes. This research not only reinforces existing evidence but also, for the first time, provides theoretical support for the

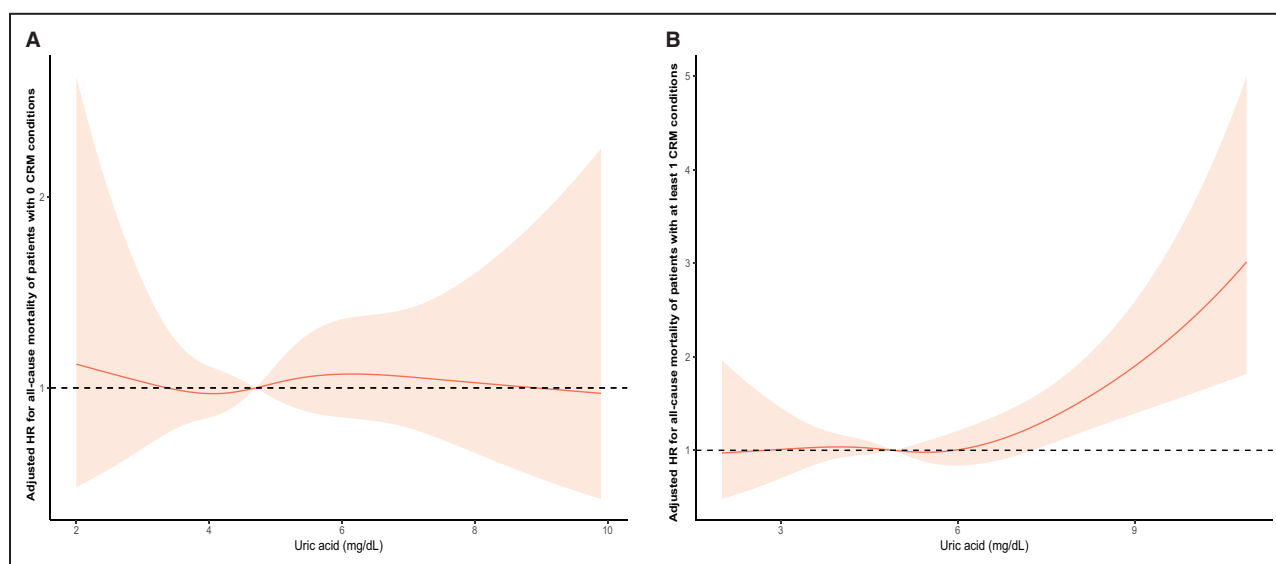


Figure 2. Dose-response relationship between SUA level (mg/dL) and the risk of all-cause mortality in participants with different CRM conditions from CHARLS.

A, 0 CRM conditions. B, ≥1 CRM condition. Data were adjusted for age, sex, residence, education, body mass index, hypertension, smoking status, estimated glomerular filtration rate, and triglyceride. CHARLS, China Health and Retirement Longitudinal Study; CRM, cardiac, renal, and metabolic; and HR, hazard ratio.

management of hyperuricemia in patients with CRM conditions using representative cohort data. It underscores the clinical importance of effectively managing SUA and gout to improve health outcomes in middle-aged and older populations, offering critical insights for optimizing clinical practice.

Some observational studies have shown that elevated SUA levels are an independent risk factor for CVD, including hypertension, coronary artery disease, atrial fibrillation, and heart failure.^{26,27} Research suggests that hyperuricemia promotes the onset and progression of CVD through several mechanisms. First, hyperuricemia activates the renin-angiotensin system, leading to vasoconstriction and sodium retention, which contribute to the development of hypertension.²⁸ Second, urate salts participate in vascular inflammation and oxidative stress by increasing the oxidation of low-density lipoprotein, lipid peroxidation, and C-reactive protein levels, thereby damaging endothelial cells and promoting the progression of hypertension.²⁹ Additionally, urate salts activate the mitogen-activated protein kinase pathway, inducing cyclooxygenase-2 expression and upregulating platelet-derived growth factor, which stimulates the proliferation and migration of vascular smooth muscle cells, ultimately leading to vascular remodeling and the progression of CVD.^{30,31} In kidney disease, the deposition of urate salts in the renal tubules and interstitium significantly inhibits the expression of nitric oxide synthase in the kidney. The reduction in nitric oxide disrupts the regulation of renal vascular tone, renal blood flow, and glomerular

feedback.³² Furthermore, urate directly affects endothelial cells by decreasing nitric oxide levels and triggering vascular smooth muscle cell proliferation, extracellular matrix deposition, and macrophage adhesion and migration.³³ These processes contribute to increased arterial resistance and renal dysfunction. Moreover, hyperuricemia is closely associated with insulin resistance through mechanisms such as direct cytotoxic effects on pancreatic β -cells, reduced insulin secretion, and decreased insulin receptor sensitivity. Insulin resistance, in turn, exacerbates hyperuricemia by upregulating the expression of URAT1 (urate transporter 1) in the renal cortex.^{34,35} It is noteworthy that gout is a stronger risk factor for mortality than isolated hyperuricemia. Previous clinical studies on gout have also shown a high prevalence of chronic comorbidities, consistent with the findings of this study.^{36–38} Gout, characterized by monosodium urate crystal deposition in joints, represents a clinical manifestation of elevated SUA levels. In addition to the previously discussed role of SUA in promoting CRM, the inflammatory response associated with gout constitutes another critical mechanism contributing to CRM. Monosodium urate crystal deposition induces robust inflammatory responses through direct neutrophil activation and the NACHT-LRR-PYD-containing protein 3 inflammasome pathway, resulting in the release of proinflammatory cytokines during acute gout attacks.³⁹ Recent evidence suggests that this inflammation persists between attacks, with ultrasound studies confirming ongoing synovitis and inflammation.⁴⁰ Chronic inflammation in

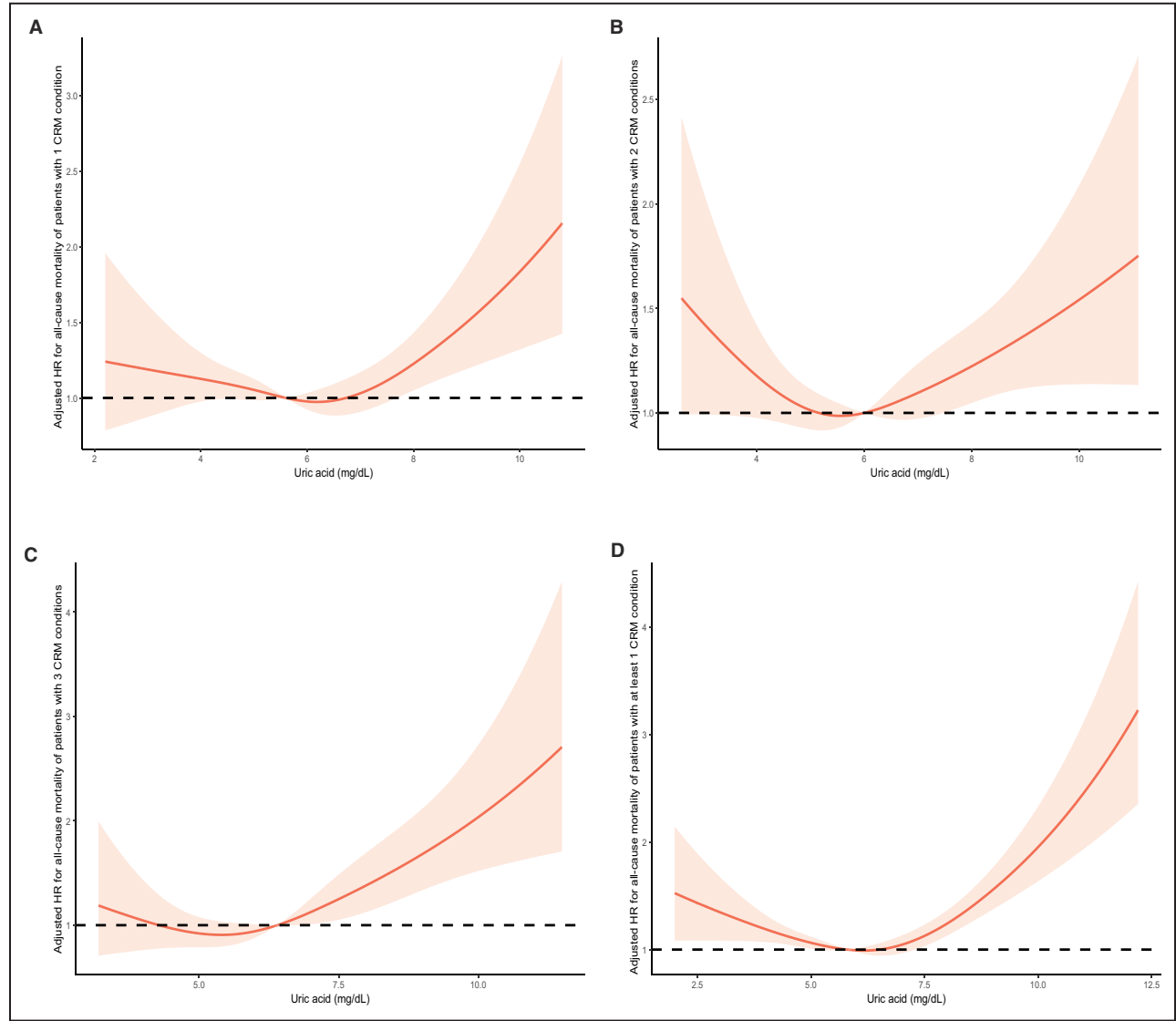


Figure 3. Dose–response relationship between SUA level (mg/dL) and the risk of all-cause mortality in participants with different CRM conditions from NHANES. **A**, 1 CRM condition. **B**, 2 CRM conditions. **C**, 3 CRM conditions. **D**, ≥1 CRM condition. Data were adjusted for age, sex, race, education, poverty, body mass index, hypertension, smoking status, estimated glomerular filtration rate, and triglyceride. CRM indicates cardiac, renal, and metabolic; HR, hazard ratio; and NHANES, National Health and Nutrition Examination Survey.

gout extends its impact beyond atherosclerosis and CVD, playing a pivotal role in the pathogenesis of other conditions such as diabetes and CKD.^{36,41}

There were also some conflicting results. A study conducted in Taiwan found that individuals with SUA levels at either extreme were at higher risk of all-cause

Table 5. MR Estimates From Different Methods of Assessing the Causal Effect Between SUA and Each CRM Condition

	No. of SNVs	IVW		MR-Egger		Weighted median		MR-PRESSO	
		OR (95% CI)	P value	OR (95% CI)	P value	OR (95% CI)	P value	OR (95% CI)	P value
SUA on CVD	148	1.109 (1.024–1.201)	0.01	1.118 (0.950–1.316)	0.18	1.151 (1.040–1.273)	<0.01	1.124 (1.048–1.206)	<0.01
SUA on CKD	149	1.441 (1.231–1.688)	<0.01	1.146 (0.834–1.575)	0.403	1.391 (1.053–1.837)	0.02	1.425 (1.219–1.665)	<0.01
SUA on diabetes	149	1.189 (1.075–1.316)	<0.01	1.200 (0.975–1.476)	0.09	1.173 (1.024–1.344)	0.02	1.171 (1.063–1.289)	<0.01

CKD indicates chronic kidney disease; CRM, cardiac, renal, and metabolic; CVD, cardiovascular disease; IVW, inverse variance weighting; MR, Mendelian randomization; MR-PRESSO, Mendelian Randomization Pleiotropy Residual Sum and Outlier; OR, odds ratios; SNV, single-nucleotide variant; and SUA, serum uric acid.

and cardiovascular mortality.⁴² Another systematic review suggested that urate-lowering therapy did not improve all-cause mortality or prevent renal failure.⁴³ In our study, discrepancies were observed in the stratified analyses across the 2 data sets. Differences in population demographics, ethnicities, or adjusted confounding factors may have contributed to the observed disparities in the results. To better address these issues, we used MR analysis to establish the causal relationship between SUA and CVD, CKD, and diabetes. By effectively eliminating confounding factors, such as environmental influences, MR provides more reliable causal inferences, helping to overcome the limitations of traditional observational studies.

Several limitations should be considered when interpreting our findings. The 2 study samples were from different countries, which allowed us to explore the relationship between uric acid and CRM conditions in diverse ethnic groups. The distinct origins of Sample 1 and Sample 2, as well as racial, age, and sex variations in the distribution of these samples, might have led to some disparities, partly explaining the different results observed in the stratified analyses. Despite these variations, the primary findings across both samples were consistent, showing a significant association between SUA and an increased prevalence of CRM conditions and all-cause mortality in patients with CRM conditions. This consistency across diverse samples strengthens the reliability of our conclusions. Data availability prevented us from including all potential confounding factors. Due to the limited number of outcome events in some subgroups, we were able to conduct stratified analyses only in participants with 0 or ≥ 1 CRM condition. The participants in our MR analyses were primarily of European ancestry. Although minimizing the potential bias from population stratification, this may not fully represent other ethnic groups.

Conclusions

In conclusion, combining 2 large data sets and MR analysis enabled us to find that SUA and gout are both strong risk factors for the prevalence and all-cause mortality of CRM conditions. Our results underscore the importance of managing hyperuricemia and gout in patients with CRM conditions. Causal effects of SUA genetic variants on CRM conditions were confirmed through MR analyses.

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Disclosures

None.

Supplemental Material

Data S1.

Tables S1–S11.

Figures S1–S15.

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